

Lab Report: Comparison of Virtual Machines (VMs) and Containers

1. Objective

The primary goals of this experiment are:

- To understand the conceptual and practical differences between Virtual Machines (VMs) and Containers.
 - To install and configure an Ubuntu-based Nginx web server using both VirtualBox/Vagrant and Docker inside WSL.
 - To compare resource utilization, performance, and operational characteristics of both environments.
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2. Hardware and Software Requirements

Hardware

- 64-bit system with virtualization support enabled in BIOS.
- Minimum 8 GB RAM (4 GB acceptable).
- Internet connection.

Software

- Oracle VirtualBox and Vagrant.
 - Windows Subsystem for Linux (WSL 2) with Ubuntu distribution.
 - Docker Engine (docker.io).
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3. Theory

Feature	Virtual Machine (VM)	Container
Virtualization Level	Emulates complete hardware and kernel.	Virtualizes at the OS level, sharing the host kernel.
Isolation	Strong (Full OS isolation).	Moderate (Process-level isolation).
Resource Usage	Higher (Requires dedicated RAM/CPU for guest OS).	Lightweight and efficient.
Startup Time	Slower (Minutes/Seconds).	Fast (Milliseconds/Seconds).

4. Experiment Setup - Part A: Virtual Machine

Step 1: Initialization and Deployment

Using Vagrant, an Ubuntu VM was initialized and started.

- **Command:** `vagrant init ubuntu/jammy64` followed by `vagrant up`.

```
PS C:\Users\Saharsh Kumar> mkdir vm-lab

Directory: C:\Users\Saharsh Kumar

Mode                LastWriteTime         Length Name
----                -
d-----          31-01-2026    23:15             vm-lab

PS C:\Users\Saharsh Kumar> cd vm-lab
PS C:\Users\Saharsh Kumar\vm-lab> vagrant init ubuntu/jammy64
A 'Vagrantfile' has been placed in this directory. You are now
ready to 'vagrant up' your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
'vagrantup.com' for more information on using Vagrant.
PS C:\Users\Saharsh Kumar\vm-lab> vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Importing base box 'ubuntu/jammy64'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'ubuntu/jammy64' version '20241002.0.0' is up to date...
==> default: Setting the name of the VM: vm-lab_default_1769882140559_18457
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
        default: Adapter 1: nat
==> default: Forwarding ports...
        default: 22 (guest) => 2222 (host) (adapter 1)
```

Observation: The system downloads the base box (Ubuntu Jammy) and configures the VirtualBox provider. Port forwarding (2222 -> 22) is established.

Step 2: Accessing the VM (SSH)

Once the VM was up, we established a connection to the guest OS.

- **Command:** `vagrant ssh`

```
PS C:\Users\Saharsh Kumar\vm-lab> vagrant ssh
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-164-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 31 18:08:05 UTC 2026

System load:            0.55
Usage of /:              4.5% of 38.70GB
Memory usage:           32%
Swap usage:              0%
Processes:              111
Users logged in:         0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd00::28:41ff:fee4:55a4

Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
```

Observation: Successful login to the Ubuntu 22.04.5 LTS environment.

Step 3: Installing Nginx

Inside the VM terminal, the package lists were updated, and the Nginx web server was installed.

- **Commands:** `sudo apt update`, `sudo apt install -y nginx`

```
vagrant@ubuntu-jammy:~$ sudo apt update
sudo apt install -y nginx
sudo systemctl start nginx
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 2439 (unattended-upgr)...
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 2439 (unattended-upgr)...
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 2439 (unattended-upgr)...
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Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 2439 (unattended-upgr)...
```

Observation: The apt package manager retrieves necessary archives. This process is slower than Docker as it installs dependencies for a full OS environment.

Step 4: Verification Inside VM

We verified the server was running locally within the guest OS.

- **Command:** `curl localhost`

```
vagrant@ubuntu-jammy:~$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Observation: The curl command inside the VM returns the full HTML source of the "Welcome to nginx!" page.

5. Experiment Setup - Part B: Containers (Docker)

Step 1: Running the Container

The Docker engine was used to pull the Ubuntu image and deploy a containerized Nginx instance.

- **Command:** `docker run -dp 8080:80 --name nginx-container nginx`

```
PS C:\Users\Saharsh Kumar\vm-lab> wsl
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
a3629ac5b9f4: Pull complete
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker run -d -p 8080:80 --name nginx-container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
70014b6c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
2b4aca5cf3b2631be5b89041c077ce31f7db9189fc06dea8f7e47b15be962c45
```

Observation: Docker pulls the image layers and starts the container nearly instantaneously.

Step 2: Verification

The Nginx server was verified by accessing the mapped port on the localhost.

- **Command:** `curl localhost:8080`

```
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Observation: The curl command confirms the Nginx "Welcome" page is active on port 8080.

6. Resource Utilization & Comparison

This section uses specific metrics captured during the experiment to contrast the two technologies.

A. Boot Time Analysis

- **Metric:** Time taken to reach a usable state.
- **VM Command:** systemd-analyze

```
vagrant@ubuntu-jammy:~$ systemd-analyze
Startup finished in 6.948s (kernel) + 29.871s (userspace) = 36.819s
graphical.target reached after 27.730s in userspace
```

Observation (VM): The VM took **36.819 seconds** to finish startup (6.9s kernel + 29.8s userspace). **Observation (Container):** The container started in **less than 1 second** (refer to Docker output in Sec 5).

B. Process Overhead & Isolation

- **Metric:** Number of background processes required to run the application.
- **VM Command:** htop

0%	0.0%	Tasks: 29, 25 thr; 1 running															
1%	0.0%	Load average: 0.01 0.18 0.16															
Mem[]	194M/957M	Uptime: 00:09:43															
Swap[]	0K/0K																
PID	USER	PRI	NI	VR	RES	SHR	S	CP	PR	MEM	TIME	Command					
689	syslog	20	0	217M	5344	4272	S	0.0	0.5	0:00.04		/usr/sbin/rsyslogd -n -iNONE					
690	root	20	0	82700	3876	3536	S	0.0	0.4	0:00.00		/usr/sbin/irqbalance --foreground					
692	root	20	0	1367M	37756	23944	S	0.0	3.9	0:02.13		/usr/lib/snapd/snapd					
694	root	20	0	23696	7200	6240	S	0.0	0.7	0:00.20		/lib/systemd/systemd-logind					
696	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.55		/usr/lib/snapd/snapd					
697	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.19		/usr/lib/snapd/snapd					
698	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.00		/usr/lib/snapd/snapd					
699	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.00		/usr/lib/snapd/snapd					
705	syslog	20	0	217M	5344	4272	S	0.0	0.5	0:00.01		/usr/sbin/rsyslogd -n -iNONE					
706	syslog	20	0	217M	5344	4272	S	0.0	0.5	0:00.00		/usr/sbin/rsyslogd -n -iNONE					
707	syslog	20	0	217M	5344	4272	S	0.0	0.5	0:00.00		/usr/sbin/rsyslogd -n -iNONE					
736	root	20	0	6220	1108	1020	S	0.0	0.1	0:00.02		/sbin/agetty -o -p -- \u --keep-baud 115200,57600,38400,9600 ttyS0 vt220					
750	root	20	0	6176	1104	1016	S	0.0	0.1	0:00.03		/sbin/agetty -o -p -- \u --noclear ttyL linux					
761	root	20	0	107M	21312	13192	S	0.0	2.2	0:00.20		/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-sign					
769	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.08		/usr/lib/snapd/snapd					
770	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.20		/usr/lib/snapd/snapd					
771	root	20	0	107M	21312	13192	S	0.0	2.2	0:00.00		/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-sign					
783	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.25		/usr/lib/snapd/snapd					
842	root	20	0	15440	8880	7248	S	0.0	0.9	0:00.04		sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups					
856	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.26		/usr/lib/snapd/snapd					
862	root	20	0	1367M	37756	23944	S	0.0	3.9	0:00.24		/usr/lib/snapd/snapd					
1881	root	20	0	16920	9868	7716	S	0.0	1.0	0:00.03		sshd: vagrant [priv]					
1884	vagrant	20	0	17076	9316	7724	S	0.0	1.0	0:00.10		/lib/systemd/systemd --user					
1885	vagrant	20	0	102M	4780	0	S	0.0	0.5	0:00.00		(sd-pam)					
1948	vagrant	20	0	17472	7824	5300	S	0.0	0.8	0:00.50		sshd: vagrant@pts/0					
1949	vagrant	20	0	9156	5016	3272	S	0.0	0.5	0:00.05		-bash					
2271	root	20	0	289M	19444	16524	S	0.0	2.0	0:00.07		/usr/libexec/packagekitd					
2272	root	20	0	289M	19444	16524	S	0.0	2.0	0:00.00		/usr/libexec/packagekitd					
2273	root	20	0	289M	19444	16524	S	0.0	2.0	0:00.00		/usr/libexec/packagekitd					
2275	root	20	0	229M	6804	6176	S	0.0	0.7	0:00.02		/usr/libexec/polkitd --no-debug					
2276	root	20	0	229M	6804	6176	S	0.0	0.7	0:00.00		/usr/libexec/polkitd --no-debug					
2278	root	20	0	229M	6804	6176	S	0.0	0.7	0:00.00		/usr/libexec/polkitd --no-debug					
2824	root	20	0	55200	12020	10428	S	0.0	1.2	0:00.02		nginx: master process /usr/sbin/nginx -g daemon on; master_process on;					
2827	www-data	20	0	55872	5400	3484	S	0.0	0.6	0:00.00		nginx: worker process					
2828	www-data	20	0	55872	5400	3484	S	0.0	0.6	0:00.00		nginx: worker process					
2931	vagrant	20	0	8708	4468	3412	R	0.0	0.5	0:00.10		htop					
F1	Help	F2	Setup	F3	Search	F4	Find	F5	SortBy	F6	Nice	F7	Nice	F8	Nice	F9	Quit

Observation (VM): htop reveals a heavy process tree. Even though we only want Nginx, the VM is running systemd, snapd, rsyslogd, polkitd, and sshd. There are dozens of tasks running to support the OS.

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
aca20b894db3	nginx-container	0.00%	13.22MiB / 7.592GiB	0.17%	2.02kB / 1.4kB	0B / 12.3kB	17

Observation (Container): docker stats shows the container uses minimal resources because it *only* runs the application process (Nginx) and its direct dependencies.

C. Memory Usage

- **Metric:** RAM consumption.

```
</html>
vagrant@ubuntu-jammy:~$ free -h
             total        used        free      shared  buff/cache   available
Mem:          957Mi       196Mi       171Mi         0Ki        589Mi        604Mi
Swap:           0B           0B           0B
Startup finished in 6.948s (kernel) + 29.871s (userspace) = 36.819s
graphical.target reached after 27.730s in userspace
```

Observation (VM): The free -h command inside the VM shows it has allocated **957Mi** total, with **196Mi** used immediately by the OS kernel and services.

Observation (Container): Referring to the Docker Stats image above, the container consumes only **13.22MiB** of RAM.

Comparison Summary Table

Parameter	Virtual Machine (VM)	Container (Docker)
Boot Time	~36.8 seconds (Measured)	< 1 second
RAM Usage	~196 MiB (Base OS overhead)	~13.22 MiB (App only)
Background Processes	High (init, systemd, ssh, logs)	Low (only Nginx entrypoint)
Disk Usage	Larger (Full OS libraries)	Smaller (Layered Image)

7. Conclusion

The experiment validates that **Containers** are significantly more lightweight.

1. **Speed:** The Docker container started instantly, whereas the VM required nearly 37 seconds to boot.
2. **Efficiency:** The VM required ~200MB of RAM just to exist, while the containerized application ran comfortably on ~13MB.
3. **Complexity:** The htop analysis clearly shows that VMs run a full operating system stack, creating unnecessary overhead for single-application deployments.

Therefore, Containers are ideal for microservices and rapid scaling, while VMs are better suited for scenarios requiring full hardware simulation or complete OS isolation.